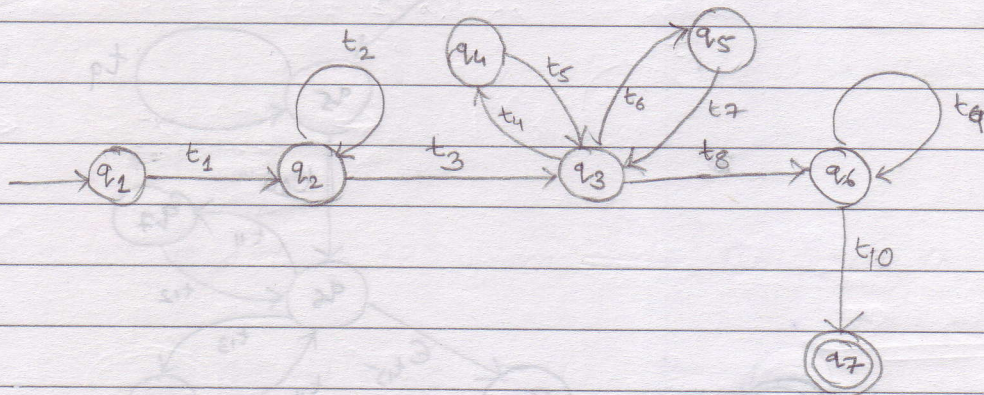


FOUNDATIONS OF COMPUTING THEORY - HOMEWORK 07

Q.1) a) 2-PDA for $\{w\#w\# \mid w \in \{a, b\}^*\}$



$$t_1 = S_1: \epsilon, \epsilon \rightarrow \$, S_2: \epsilon, \epsilon \rightarrow \epsilon$$

$$t_2 = S_1: a, \Gamma \rightarrow a, \text{ where } \Gamma = \{\$, a, b\}$$

$$S_1: b, \Gamma \rightarrow b,$$

$$S_2: a, \epsilon \rightarrow \epsilon, S_2: b, \epsilon \rightarrow \epsilon$$

$$t_3 = S_1: \#, \epsilon \rightarrow \epsilon, S_2: \#, \epsilon \rightarrow \epsilon$$

$$t_4 = S_1: \epsilon, a \rightarrow \epsilon, S_2: \epsilon, \epsilon \rightarrow \epsilon$$

$$t_5 = S_1: \epsilon, \epsilon \rightarrow \epsilon, S_2: \epsilon, \epsilon \rightarrow a$$

$$t_6 = S_1: \epsilon, b \rightarrow \epsilon, S_2: \epsilon, \epsilon \rightarrow \epsilon$$

$$t_7 = S_1: \epsilon, \epsilon \rightarrow \epsilon, S_2: \epsilon, \epsilon \rightarrow b$$

$$t_8 = S_1: \epsilon, \$ \rightarrow \epsilon, S_2: \epsilon, \epsilon \rightarrow \epsilon$$

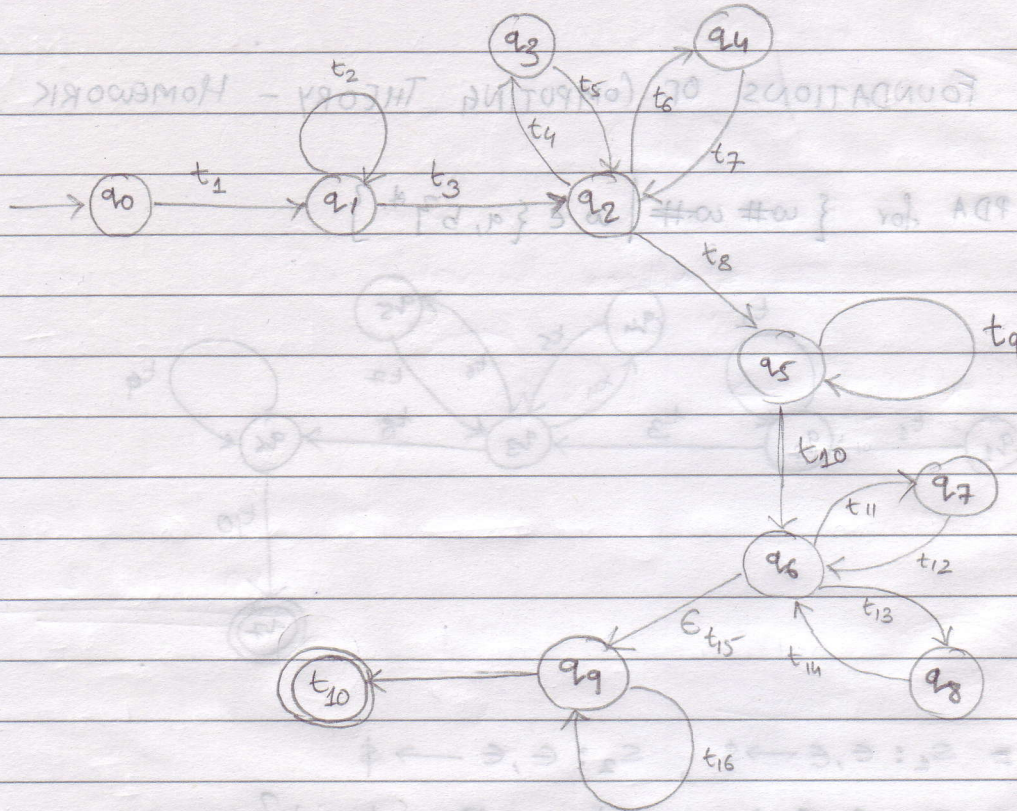
$$t_9 = S_1: a, a \rightarrow a, S_2: b, b \rightarrow b$$

$$S_1: a, \epsilon \rightarrow \epsilon, S_1: a, \epsilon \rightarrow \epsilon$$

$$t_{10} = S_1: \#, \$ \rightarrow \epsilon, S_2: \#, \$ \rightarrow \epsilon$$

- The transitions are labelled as $S_i: \epsilon, \Gamma_\epsilon \rightarrow \Gamma_\epsilon$ where S_i is the stack on which we do the remaining part as we would do in a PDA.

Q.1 b)



t_1 to t_8 is the same Q.1 a)

$t_9 = S_1: a, \Gamma \rightarrow a, S_1: b, \Gamma \rightarrow b$

$S_2: a, a \rightarrow \epsilon, S_2: a, a \rightarrow \epsilon$

$t_{10} = t_3$

$t_{11} = t_4, t_{12} = t_5, t_{13} = t_6, t_{14} = t_7$

$t_{15} = t_8$

$t_{16} = t_9$

$t_{17} = t_{10}$

Q.1 c) Turing-recognizable languages.

Q.2. a) $T = (Q, \Sigma, \Gamma, \delta, q_{\text{accept}}, q_{\text{reject}})$

$T' = (Q', \Sigma', \Gamma', \delta', q'_{\text{accept}}, q'_{\text{reject}})$

$$q' \neq q$$

$$\epsilon' = \epsilon$$

$$q'_{\text{accept}} = q_{\text{reject}}$$

$$q'_{\text{reject}} = q_{\text{accept}}$$

Q.2. b) If T is not a decider and we give T a string $\epsilon \in L(T)$ then maybe on a given string, T will ~~either~~ neither. Then on some strings, T will neither accept nor reject. Since T and T' is the same but with the q_{accept} and q_{reject} states swapped, T' will also neither accept nor reject.
 - Hence neither of the TM will accept or reject on the same of the input strings.

Q.4) a) $A = \{ \langle D \rangle \mid D \text{ is a DFA that accepts } \Sigma^* \}$

Q.4) b) * $S =$ "On input $\langle D \rangle$ where D is a DFA

1. Swap the accept and reject states of D (say $\bar{D} = c$)
2. Run TM M from Theorem 4.4 on c .
3. If M accepts, accept; if M rejects, rejects."

Q.5) $S =$ "On input $\langle A, B \rangle$ where A and B are NFAs

1. Swap the accept and reject states of B (say $\bar{B} = c$)
2. Construct DFA D using subset construction such that

$$\bar{D} = A \quad L(D) = L(A \cap \bar{B})$$

$$L(D) = L(A) \cap \overline{L(B)}$$

3. Run TM M from Theorem 4.4 on D

4. If M accepts, accept; if M rejects, rejects."

Q.3) a) Machine M_1 and M_2 are run one after the other
 $\therefore$ If M_1 goes into a loop, the string is never simulated on M_2 . As a result we cannot handle the cases where both go into a loop and when either one goes into a loop.
 $\therefore$ The statement does not show that RE is closed under union.

Q.3. b) Part a) does not ~~show~~ imply RE is not closed under union because

- If we simulate T_1 and T_2 simultaneously for cases as listed below

T_1	T_2
Accepts/rejects	Accepts/rejects

Infinite loop	Accepts/rejects
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Accepts/rejects	Infinite loop
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Infinite loop	Infinite loop
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are all handled. Hence it does not imply RE not closed under union.